

Akash Gupta

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📺 ConceptsOfAI

🎓 Education

M.Tech (Artificial Intelligence)
IIT Kharagpur
07/2024 – present

B.Tech
IIT (BHU), Varanasi
07/2012 – 05/2016

Higher secondary, IL school
2011– 85.6%

🏆 Accomplishment

IIT JEE 2012
All India Rank (AIR) -2954

Actuarial Statistics (CS1) by IFoA, UK
2023
Cleared Actuarial Statistics (CS1) exam

🧠 Skills

Programming & Databases ● ● ● ● ●
Python, SQL

LLM ● ● ● ● ●
Langchain, LlanGgraph

Git ● ● ● ● ●

ML, DL ● ● ● ● ●
PyTorch, TensorFlow

Data Visualization & Analytics: ● ● ● ● ●
Tableau

Web & API Development: ● ● ● ● ●
FastAPI, Flask

Cloud & Deployment ● ● ● ● ●
AWS, Supabase, Docker

N8N ● ● ● ● ●

🧩 Technical Initiative

Concepts of AI (YouTube) [🔗](#)
Simplifying AI/ML concepts

👤 Profile

Enthusiastic and detail-oriented aspiring Data Scientist with strong foundational knowledge in machine learning, data analysis, and statistical modeling. Proficient in Python, SQL, and TensorFlow, with hands-on experience gained through academic projects, internships, or coursework. Passionate about solving complex problems and eager to apply technical skills to drive innovation and insights in a dynamic organization.

💼 Professional Journey

YAL [🔗](#)
Jr AI Scientist
03/2026 – Present

- Working in NLP division of AI: Research and Innovation team to build AI powered scam-free, privacy-first communication system
- Developing an on-device Smart Reply system to enable instant, privacy-preserving response suggestions for chat communication

FxisAI
Machine Learning Intern
05/2025 – 07/2025 | Remote

- Built and optimized a **Retrieval-Augmented Generation (RAG) pipeline** using Weaviate at semantic search, text chunking, and context retrieval to improve query response accuracy.
- Developed an **AI-driven financial data extraction system**, leveraging OCR and regex/LLMs to parse unstructured invoices into structured formats (Excel/DataFrames) for downstream analytics.
- Gained **exposure to AI consultancy practices** by engaging with real-world problem statements, understanding client requirements, and observing how technical solutions are aligned with business needs.

Tutela Education Private limited
Manager- IB division, Data Science educator
11/2019 – 06/2024 | Gurugram, India

- Headed the International Baccalaureate (IB) division and managed a team of 4 members
- Improved productivity by 10-20% by Streamlining course content and schedules.
- Collaborated closely with stakeholders to ensure a seamless journey for clients
- Started an AP Statistics course from scratch and developed its content and 70-80% students achieved highest grade in the 3 years

Volvo Eicher Commercial vehicles
Graduate Engineer Trainee
06/2016 – 04/2017 | Indore, India

- Worked in Strategic sourcing department and collaborated with cross-functional teams to derive insights from technical product data and commercial viability reports.
- Gained experience in project coordination and using data-driven methods to make business decisions related to strategic sourcing and supplier management
- Assisted in developing cost quotations and proposals for new parts
- Managed workflow optimization on the assembly line

📁 Projects

High-Throughput LLM Inference Engine (Code) [🔗](#)

- Built a **self-hosted LLM inference server** using **vLLM + FastAPI** with async request queuing and **continuous batching (PagedAttention)**, replacing costly OpenAI API dependency with on-premise GPU serving (**Demo Video**) [🔗](#)
- Load tested with **50 concurrent users** — achieved **7.5 RPS, 2.6s median latency, 0% failure rate**; containerized with **Docker** for production deployment (**Report**) [🔗](#)

End-to-End Deployed Financial LLM Inference System (Code) [🔗](#)

- Built and deployed** an end-to-end **Financial LLM inference pipeline** by fine-tuning **Qwen-2 (0.5B)** on a proprietary financial instruction dataset using **LoRA**, followed by weight merging for standalone inference (**Demo Video**) [🔗](#).
- Dockerized** and productionized a **FastAPI-based CPU inference** service, deployed on **Railway**, handling real-world issues like cold starts, dependency resolution, model loading failures, and runtime crashes (**Report**) [🔗](#).
- Analyzed and optimized **inference latency (≈5–20s for 100–250 tokens)** breaking it into **prefill vs decode** phases, reducing prompt/tokenization overhead, enforcing deterministic decoding, and experimenting with CPU **quantization**, and optimisation

Freelancer Project Price Estimator (code link) [🔗](#)

- Built a Deep Learning-based Freelance **Project Price Estimator**, leveraging trained embeddings and multi-input **neural network** to predict project budgets, deployed as an interactive **web app (Demo Video)** [🔗](#)

- Engineered dataset of ~9k freelance postings, applied **text embeddings, and log-transformed regression** to handle skewed price distributions and outliers, improving prediction robustness(**Report Link**)  .

AI powered Resume–Job Description Matcher (Code Link)

- Built an Intelligent Resume–Job Description Matcher using SBERT/CodeBERT/GraphCodeBERT with skills ontology and regex parsing for semantic alignment between resumes and JDs(**Demo Video**) 
- Designed a multi-criteria scoring system (skills, projects, experience, qualifications) with dual persona AI reports for candidates & HR, and deployed the end-to-end system with **Flask on Render (Report Link)** 

Banking Complaint Severity Classifier(Code Link)

- Developed a **DistilBERT-based text classification system** fine-tuned with **LoRA** achieving **~76% accuracy** for classifying complaints into severity levels (**Demo Video**) 
- Built an interactive **Flask web app** with stylish **HTML/CSS frontend** with custom **multi-threshold severity mapping**, and real-time predictions (**Report Link**) 

AI-Powered Legal Document RAG Bot (Code Link)

- Built an AI-Powered Legal Document RAG Bot for querying legal/constitutional documents using **BM25** + dense retrievers (**ChromaDB, MiniLM**) with query transformation, conversational memory, and prompt engineering for structured responses (**Demo Video**) 
- Deployed the chatbot as a **Flask web app** hosted on **Render** for interactive use (**Report Link**) 